

Computational Thermal Engineering Day – 10

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Create a surface file “Sphere”

1. Use modelling software like FreeCAD to model sphere
2. Sphere dimensions are as follows:
 - i. Radius = 0.1mm
 - ii. Location of the sphere = (0.5mm, 0.5mm, 0.5mm)
3. Export the sphere file with the file name, “sphere.obj”

Create a case setup file for the problem, “Flow over a sphere”

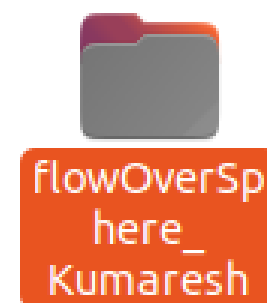
Use **icoFoam** solver

From the location:

```
openfoam@openfoam: ~/OpenFOAM/openfoam/tutorials/in...
openfoam@openfoam:~/OpenFOAM/openfoam/tutorials/incompressible/icoFoam/cavity$
```

Copy 0, constant, and system files
in the new file name

“flowOverSphere_YourNAME”



Name
0
constant
system

Modify the case setup file

Changes in system/**blockMesh** file

icoFoam file

flowOverSphere

```
scale 0.1;

vertices
(
    (0 0 0)
    (1 0 0)
    (1 1 0)
    (0 1 0)
    (0 0 0.1)
    (1 0 0.1)
    (1 1 0.1)
    (0 1 0.1)
);

blocks
(
    hex (0 1 2 3 4 5 6 7) (20 20 1) simpleGrading (1 1 1)
);
```



```
scale 1;

vertices
(
    (0 0 0)
    (1 0 0)
    (1 1 0)
    (0 1 0)
    (0 0 1)
    (1 0 1)
    (1 1 1)
    (0 1 1)
);

blocks
(
    hex (0 1 2 3 4 5 6 7) (20 20 20) simpleGrading (1 1 1)
);
```

Modify the case setup file

Changes in system/**blockMesh** file

icoFoam file

```
boundary
(
    movingWall
    {
        type wall;
        faces
        (
            (3 7 6 2)
        );
    }
    fixedWalls
    {
        type wall;
        faces
        (
            (0 4 7 3)
            (2 6 5 1)
            (1 5 4 0)
        );
    }
    frontAndBack
    {
        type empty;
        faces
        (
            (0 3 2 1)
            (4 5 6 7)
        );
    }
);
```



flowOverSphere

```
front
{
    type wall;
    faces
    (
        (4 5 6 7)
    );
}
back
{
    type wall;
    faces
    (
        (0 3 2 1)
    );
}
```

```
boundary
(
    top
    {
        type wall;
        faces
        (
            (3 7 6 2)
        );
    }
    bottom
    {
        type wall;
        faces
        (
            (1 5 4 0)
        );
    }
    left
    {
        type patch;
        faces
        (
            (0 4 7 3)
        );
    }
    right
    {
        type patch;
        faces
        (
            (2 6 5 1)
        );
    }
}
```


Modify the case setup file

Changes in 0/p boundary file

icoFoam file

```
FoamFile
{
    version      2.0;
    format       ascii;
    class        volScalarField;
    object       p;
}
// * * * * *

dimensions      [0 2 -2 0 0 0 0];
internalField    uniform 0;
boundaryField
{
    movingWall
    {
        type      zeroGradient;
    }

    fixedWalls
    {
        type      zeroGradient;
    }

    frontAndBack
    {
        type      empty;
    }
}
```



flowOverSphere

```
FoamFile
{
    version      2.0;
    format       ascii;
    class        volScalarField;
    object       p;
}
// * * * * *

dimensions      [0 2 -2 0 0 0 0];
internalField    uniform 0;
boundaryField
{
    top
    {
        type      zeroGradient;
    }

    bottom
    {
        type      zeroGradient;
    }

    left
    {
        type      zeroGradient;
        //type     fixedFluxPressure;
        //value     uniform 0;
    }

    right
    {
        type      zeroGradient;
        //type     totalPressure;
        //p0       uniform 0;
    }
}
```

```
front
{
    type      zeroGradient;
}

back
{
    type      zeroGradient;
}

sphere
{
    type      zeroGradient;
}
```

Modify the case setup file

Changes in O/U boundary file

icoFoam file

```
FoamFile
{
  version      2.0;
  format       ascii;
  class        volVectorField;
  object       U;
}
// *****

dimensions      [0 1 -1 0 0 0 0];
internalField   uniform (0 0 0);
boundaryField
{
  movingWall
  {
    type        fixedValue;
    value       uniform (1 0 0);
  }

  fixedWalls
  {
    type        noSlip;
  }

  frontAndBack
  {
    type        empty;
  }
}
```



flowOverSphere

```
FoamFile
{
  version      2.0;
  format       ascii;
  class        volVectorField;
  object       U;
}
// *****

dimensions      [0 1 -1 0 0 0 0];
internalField   uniform (1e-8 1e-8 1e-8);
boundaryField
{
  top
  {
    type        noSlip;
  }

  bottom
  {
    type        noSlip;
  }

  left
  {
    type        fixedValue;
    value       uniform (1 0 0);
  }

  right
  {
    //type        zeroGradient;
    //type        pressureInletOutletVelocity;
    //value       uniform (0.1 0 0);
    type        outletInlet;
    outletValue uniform (1 0 0);
  }
}
```

```
front
{
  type        noSlip;
}

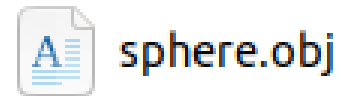
back
{
  type        noSlip;
}

sphere
{
  type        noSlip;
}
```


Modify the case setup file

Changes in constant **file**

- 1) **Create a triSurface file** by the following command:
 - **mkdir -p triSurface**
- 2) Inside the triSurface file, copy the surface file named “sphere.obj” (Carried out in step 1)



Modify the case setup file

Changes in system/**controlDict** file

icoFoam file

```
application    icoFoam;  
startFrom      startTime;  
startTime      0;  
stopAt         endTime;  
endTime        0.5;  
deltaT         0.005;  
writeControl    timeStep;  
writeInterval  20;  
purgeWrite     0;  
writeFormat    ascii;  
writePrecision 6;  
writeCompression off;  
timeFormat     general;  
timePrecision  6;  
runTimeModifiable true;
```



flowOverSphere

```
application    icoFoam;  
startFrom      startTime;  
startTime      0;  
stopAt         endTime;  
endTime        1;  
deltaT         0.0001;  
writeControl    adjustable;  
writeInterval  0.002;  
purgeWrite     0;  
writeFormat    ascii;  
writePrecision 6;  
writeCompression off;  
timeFormat     general;  
timePrecision  6;  
runTimeModifiable true;  
  
adjustTimeStep yes;  
maxCo          1;  
maxDeltaT      0.1;
```


Modify the case setup file

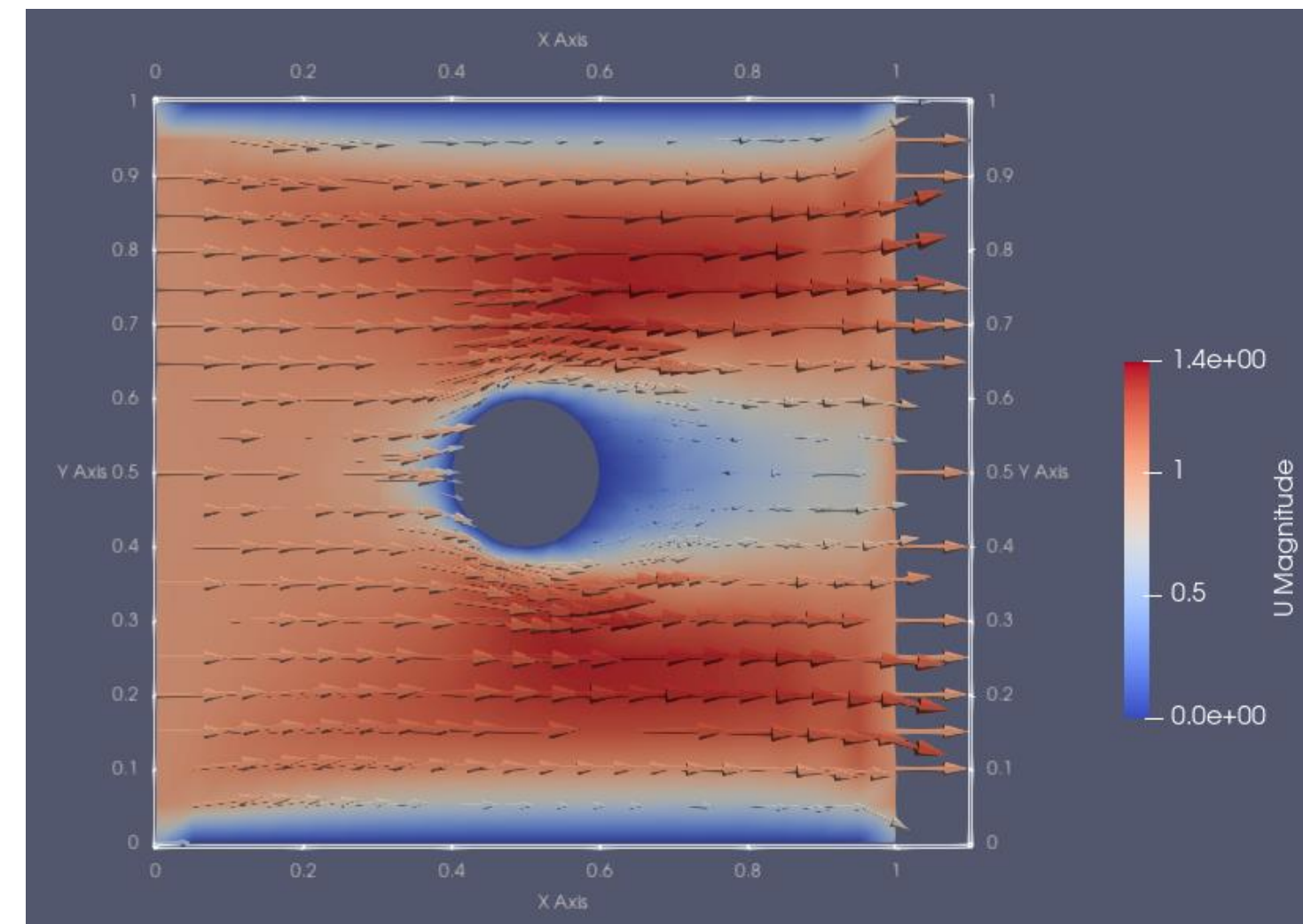
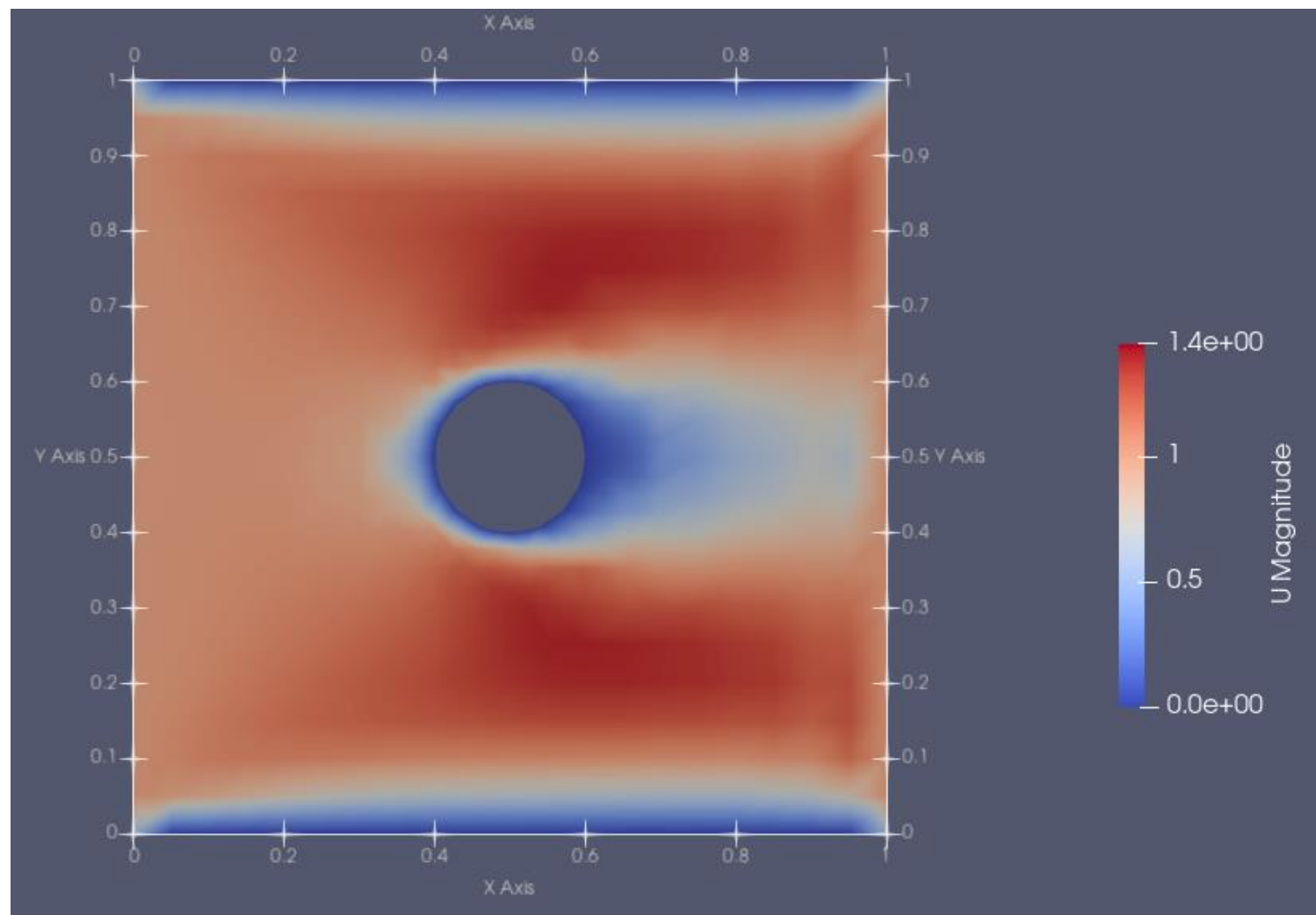
Changes in system **file**

- 1) Keep the remaining files, fvSchemes, fvSolution, and decomposeParDict as such.
- 2) Download the **following files from GitHub** and copy inside system file:
 - > meshQualityDict
 - > snappyHexMeshDict
 - > surfaceFeatureExtractDict

Commands to Run

1. blockMesh
2. surfaceFeatureExtract
3. snappyHexMesh -overwrite
4. icoFoam

Upload your new solver, case files, and results in GITHUB



THANK YOU